States of matter, atoms and the Periodic Table

Mark scheme and worked solutions for 30 questions: 10 multiple choice, 20 structured.

1 · Part 1

Paper 1 & 2 | States of matter | [1]

Answer: B

Worked solution

Learn the three descriptions together. Solid: close together, regular, vibrate about fixed positions, least energy. Liquid: close together, random, move around each other. Gas: far apart, random, move quickly in all directions, most energy.

2 · Part 1

Paper 1 & 2 | States of matter | [1]

Answer: D

Explanation

A change from solid directly to gas, without a liquid stage, is subliming (sublimation), so D is correct. A boiling and C evaporating are both changes from liquid to gas. B condensing is a change from gas to liquid.

3 · Part 1

Paper 2 only | States of matter | [1]

Answer: A

Worked solution

14.8 g dissolves in 40.0 g of water. In 1 g of water: 14.8 ÷ 40.0 = 0.370 g. In 100 g of water: 0.370 × 100 = 37.0 g. So the solubility is 37.0 g per 100 g of water.

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4 · Part 1 Paper 1 & 2 States of matter [7]

Mark scheme

(a) **[2]**

- oxygen: gas; bromine: liquid; mercury: liquid; sodium chloride: solid
- all four correct (2)
- two or three correct (1)

(b) **[1]**

- mercury (1)

(c) **[1]**

- condensing (1)

ALLOW condensation

REJECT freezing OR solidifying

(d) **[3]**

- arrangement: particles move from far apart (randomly arranged) to close together OR touching (but still randomly arranged) (1)
- movement: particles go from moving quickly in all directions to moving more slowly OR moving around each other OR sliding past each other (1)
- energy: particles have less (kinetic) energy OR particles lose energy OR energy is released to the surroundings (1)

REJECT particles stop moving

REJECT particles become regularly arranged

Worked solution

(a) Compare 25 °C with each melting and boiling point. Below the melting point a substance is solid; between the melting point and boiling point it is liquid; above the boiling point it is gas. Oxygen boils at −183 °C, so at 25 °C it is a gas. Bromine (−7 °C to 59 °C) and mercury (−39 °C to 357 °C) are liquids at 25 °C. Sodium chloride melts at 801 °C, so it is a solid.

(b) At −20 °C a substance is a liquid only if −20 °C is above its melting point and below its boiling point. Mercury (−39 °C to 357 °C) is liquid. Bromine melts at −7 °C, so at −20 °C it is still solid. Oxygen boils at −183 °C, so it is a gas, and sodium chloride is a solid.

(c) At 70 °C bromine is above its boiling point, so it is a gas. At 20 °C it is between its melting and boiling points, so it is a liquid. Gas to liquid is condensing.

(d) In the gas, particles are far apart, random and moving quickly in all directions. As the gas cools, the particles lose energy, slow down and come close together, so they are touching but still randomly arranged and able to move around each other.

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5 · Part 1 Paper 1 & 2 States of matter [7]

Mark scheme

(a) **[1]**

- diffusion (1)

(b) **[2]**

- ammonia particles OR molecules have a lower mass than hydrogen chloride particles OR molecules (1)

ALLOW ammonia has a lower Mr (17) than hydrogen chloride (36.5) OR ammonia is lighter

- so ammonia particles move OR diffuse faster and travel further (in the same time) (1)

(c) **[2]**

- gas particles collide with air particles (in the tube) (1)
- so they move in random directions OR change direction often OR zig-zag, so their overall progress along the tube is slow (1)

(d) **[2]**

- the time is shorter OR the ring forms sooner (1)
- particles have more (kinetic) energy OR move faster, so they diffuse faster (1)

IGNORE comments on the position of the ring

Worked solution

(a) Particles spreading out from where they are concentrated, by their own random movement, is diffusion.

(b) Ammonia, NH_3, has Mr = 14 + 3 = 17. Hydrogen chloride, HCl, has Mr = 1 + 35.5 = 36.5. Lighter particles move faster at the same temperature, so ammonia particles travel further before they meet the hydrogen chloride particles. The gases meet 59 cm from the ammonia end, which is closer to the hydrochloric acid end.

(c) The tube is full of air. Each gas particle keeps colliding with air particles and bouncing off in a new direction, so it moves in a random zig-zag path. Its overall movement along the tube is much slower than its actual speed.

(d) At a higher temperature the particles have more kinetic energy and move faster, so they diffuse along the tube faster and meet sooner. The ring forms in less time.

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6 · Part 1 Paper 1 & 2 States of matter [7]

Mark scheme

(a) **[1]**

- a substance that dissolves in a solvent (to form a solution) (1)

ALLOW the substance that is dissolved

(b) **[2]**

- the particles of potassium manganate(VII) (and water) move randomly OR are constantly moving (1)
- so they spread out from where they are more concentrated to where they are less concentrated OR they diffuse through the water (1)

(c) **[1]**

- 10 000 (times) (1)

ALLOW 10^4

(d) **[2]**

- there are very many particles in the crystal (1)
- the particles are very small (1)

ALLOW after many dilutions there are still enough particles to colour the solution, for 1 mark

(e) **[1]**

- a solution in which no more solute can dissolve (at that temperature) (1)

ALLOW a solution containing the maximum mass of solute that can dissolve (at that temperature)

Worked solution

(a) The solute is the substance that dissolves; here it is potassium manganate(VII). The solvent is the liquid it dissolves in (water).

(b) The particles in a liquid are always moving randomly. The purple particles spread out from the crystal, where they are concentrated, into the rest of the water, where they are less concentrated. This is diffusion, and it does not need stirring.

(c) Each step takes 10 cm^3 and makes it up to 10 + 90 = 100 cm^3, so each solution is 10 times more dilute than the one before. From solution 1 to solution 5 there are 4 dilution steps: 10 × 10 × 10 × 10 = 10 000 times more dilute.

(d) Even when the solution is 10 000 times more dilute, there are still enough purple particles to colour it. So one small crystal must contain a huge number of particles, and each particle must be extremely small.

(e) A saturated solution is one in which no more solute can dissolve at that temperature, so extra solid stays undissolved.

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7 · Part 1

Paper 2 only | States of matter | [7]

Mark scheme

(a) **[2]**

- 32 g is the maximum mass of potassium nitrate that dissolves (1)
- in 100 g of water at 20 °C (1)

ALLOW 32 g of potassium nitrate in 100 g of water at 20 °C forms a saturated solution, for 2 marks

(b) **[1]**

- 80 g (1)

(c) **[2]**

- mass that dissolves in 125 g of water = 125 × 32 ÷ 100 = 40 g (1)
- mass that does not dissolve = 50 − 40 = 10 g (1)

ECF 50 minus an incorrect mass that dissolves, for the second mark

(d) **[2]**

- the solubility of potassium nitrate increases as temperature increases OR is higher at 50 °C (1)
- at 50 °C at least 40 g dissolves in 100 g of water OR the solubility at 50 °C is at least 40 g per 100 g of water (1)

ALLOW at 50 °C all 50 g dissolves in 125 g of water, for the second mark

Worked solution

(a) Solubility is the maximum mass of a solute that will dissolve in 100 g of solvent at a stated temperature. So at 20 °C, 100 g of water can hold at most 32 g of potassium nitrate.

(b) 250 g of water is 2.5 × 100 g, so the maximum mass is 2.5 × 32 = 80 g.

(c) 125 g of water is 1.25 × 100 g, so it dissolves 1.25 × 32 = 40 g. Lena added 50 g, so 50 − 40 = 10 g stays undissolved. Check: 40 ÷ 125 × 100 = 32 g per 100 g, which matches the solubility.

(d) More solid dissolves at the higher temperature, so the solubility increases with temperature. All 50 g dissolves in 125 g of water, so at 50 °C at least 50 × 100 ÷ 125 = 40 g dissolves in 100 g of water.

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8 · Part 1

Paper 2 only | States of matter | [8]

Mark scheme

(a) **[3]**

- heat water to 40 °C and keep it at 40 °C (e.g. in a water bath, checked with a thermometer) (1)
- add potassium chloride (a little at a time) and stir (1)
- until no more dissolves OR until some solid remains undissolved (1)

ALLOW then filter (at 40 °C) to remove the undissolved solid, as an alternative for the second or third mark

(b) **[3]**

- mass of potassium chloride = 37.38 − 30.52 = 6.86 g (1)
- mass of water = 54.43 − 37.38 = 17.05 g (1)

ALLOW mass of water = 23.91 − 6.86 = 17.05 g

- solubility = 6.86 × 100 ÷ 17.05 = 40.2 (g per 100 g of water) (1)

ALLOW 40 OR 40.23

ECF from incorrect masses of potassium chloride or water, for the third mark

(c) **[2]**

- to make sure that all of the water has evaporated (1)
- if water remained, the mass of potassium chloride would be too high (and the mass of water too low) so the solubility would be too high (1)

ALLOW to make sure the result is accurate, for the second mark only if linked to removing all the water

Worked solution

(a) A saturated solution must be made at the temperature being investigated. Keep the water at 40 °C, for example in a water bath with a thermometer. Add potassium chloride with stirring until no more will dissolve and some solid is left over; then the solution is saturated.

(b) Mass of solid = (basin + dry solid) − (empty basin) = 37.38 − 30.52 = 6.86 g. Mass of water = (basin + solution) − (basin + dry solid) = 54.43 − 37.38 = 17.05 g. Solubility = 6.86 ÷ 17.05 × 100 = 40.2 g per 100 g of water. Check: mass of solution = 54.43 − 30.52 = 23.91 g, and 6.86 + 17.05 = 23.91 g.

(c) Heating to constant mass shows that no more water is being lost, so all the water has gone. Any water left behind would be counted as solid, making the mass of potassium chloride too large and the calculated solubility too high.

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9 · Part 1

Paper 1 & 2 Elements, compounds and mixtures [1]

Answer: C

Explanation

Chlorine contains only one type of atom (an element), magnesium oxide is two elements chemically combined (a compound) and sea water is a mixture of water and dissolved substances, so C is correct. A swaps the element and the compound. B and D wrongly treat sea water as a pure substance.

10 · Part 1

Paper 1 & 2 Elements, compounds and mixtures [1]

Answer: A

Explanation

Liquids that mix completely and have different boiling points are separated by fractional distillation, so A is correct. B crystallisation obtains a solid from a solution. C filtration separates an insoluble solid from a liquid. D simple distillation separates a solvent from a dissolved solid; it does not separate two liquids with boiling points this close into pure liquids.

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11 · Part 1

Paper 1 & 2 Elements, compounds and mixtures [7]

Mark scheme

(a) **[4]**

- P: element (1)
- Q: compound (1)
- R: mixture (1)
- S: element (1)

(b) **[3]**

- in the mixture, iron and sulfur are not chemically combined (1)
- so they can be separated by a physical method, such as using a magnet (1)
- in the compound (iron(II) sulfide), iron and sulfur are chemically combined OR bonded, so it has different properties from the elements, for example it is not attracted to a magnet (1)

ALLOW a compound cannot be separated by a physical method such as a magnet, for the third mark

Worked solution

(a) An element contains only one type of atom: N_2 molecules contain only nitrogen atoms, and Ne atoms are all neon. A compound contains two or more elements chemically combined: each CO_2 molecule contains carbon and oxygen bonded together. A mixture contains two or more substances that are not chemically combined: R contains nitrogen and oxygen molecules mixed together.

(b) In the mixture the iron keeps its own properties, so a magnet can pull it out. This is a physical separation, which works because the iron and sulfur are not chemically joined. After heating, a chemical reaction forms a new substance, iron(II) sulfide, in which iron and sulfur are chemically combined. The compound has its own, different properties (it is not magnetic), so a magnet cannot separate the iron from it.

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12 · Part 1

Paper 1 & 2 | Elements, compounds and mixtures | [6]

Mark scheme

(a) **[1]**

- 69 °C (1)

(b) **[1]**

- freezing (1)

ALLOW solidifying

REJECT melting OR condensing

(c) **[2]**

- the temperature stays constant at 69 °C while X changes state (1)
- a pure substance has a fixed melting (and freezing) point (1)

(d) **[2]**

- X would freeze OR solidify over a range of temperatures (1)
- the temperature would keep falling during freezing OR there would be no constant temperature section (1)

ALLOW X would start to freeze at a lower temperature, for 1 mark

Worked solution

(a) The temperature stays at 69 °C from 3 to 7 minutes while the liquid turns into a solid. A substance freezes and melts at the same temperature, so its melting point is 69 °C.

(b) X starts as a liquid and is cooling, so the change of state is freezing (liquid to solid).

(c) The temperature is exactly the same, 69 °C, for the whole time X is freezing. Pure substances melt and freeze at one fixed temperature.

(d) A mixture melts and freezes over a range of temperatures. With an impurity, the temperature would keep slowly falling while X froze, so there would be no flat, constant section.

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13 · Part 1

Paper 1 & 2 | Elements, compounds and mixtures | [6]

Mark scheme

(a) **[5]**

Indicative points, 1 mark each, max 5:

- add water to the rock salt and stir (ALLOW warm the water) to dissolve the sodium chloride (1)
- filter the mixture to remove the sand (1)
- heat the filtrate (in an evaporating basin) to evaporate some of the water (1)
- until crystals start to form OR until the solution is saturated (ALLOW leave in a warm place for the water to evaporate slowly) (1)
- leave the solution to cool so that crystals form (1)
- filter off OR decant the crystals and dry them between filter papers OR in a warm oven (1)

ALLOW heat the filtrate until about half the water has evaporated, for the third mark

REJECT filter the rock salt before adding water

(b) **[1]**

- filtrate (1)

Worked solution

(a) First add water and stir so that the sodium chloride dissolves; the sand does not dissolve. Filter the mixture: the sand stays on the filter paper as the residue, and sodium chloride solution passes through. Heat the solution in an evaporating basin to boil off some of the water until crystals begin to appear. Leave it to cool so more crystals form. Remove the crystals by filtering, and dry them between filter papers or in a warm oven.

(b) The liquid that passes through the filter paper is called the filtrate.

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14 · Part 1

Paper 1 & 2 | Elements, compounds and mixtures | [7]

Mark scheme

(a) **[4]**

- (when sea water is heated) water boils OR evaporates to form water vapour OR steam (1)
- the salt (does not evaporate because it) has a much higher boiling point, so it stays in the flask (1)
- the water vapour passes into the condenser, where it is cooled (1)
- the water vapour condenses to (pure liquid) water, which is collected in the beaker (1)

(b) **[1]**

- 100 °C (1)

(c) **[2]**

- the salt is dissolved, so it passes through the filter paper (with the water) (1)
- distillation works because water evaporates but the dissolved salt does not OR distillation separates a liquid from a dissolved solid because of the large difference in boiling points (1)

Worked solution

(a) When the sea water is heated, the water boils and becomes water vapour. The dissolved salt has a much higher boiling point, so it does not evaporate and stays behind in the flask. The vapour travels through the side arm into the condenser, where the cold water around it cools it. The vapour condenses back into liquid water, which drips into the beaker. This liquid is pure water.

(b) The thermometer measures the temperature of the vapour going into the condenser. Pure water boils at 100 °C.

(c) Filtration only separates an insoluble solid from a liquid. Salt is dissolved in sea water, so its particles pass straight through the filter paper with the water. Distillation can separate them because the water turns to vapour and leaves while the salt stays in the flask.

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15 · Part 1

Paper 1 & 2 Elements, compounds and mixtures [7]

Mark scheme

(a) **[3]**

- spot 1: 1.5 ÷ 7.5 = 0.20 (1)
- spot 2: 4.1 ÷ 7.5 = 0.55 (1)

ALLOW 0.547 OR 0.5467

- spot 3: 6.3 ÷ 7.5 = 0.84 (1)

ALLOW 0.2 for spot 1

REJECT values with a unit

(b) **[2]**

- P, R and T (2)
- any two of P, R and T with no incorrect colouring (1)

ECF colourings matching their Rf values from (a)

(c) **[2]**

- the Rf value of a substance depends on the solvent used (1)
- so in ethanol the colourings would have different Rf values from those in the book, and matching them would not identify the colourings (1)

Worked solution

(a) Rf = distance moved by the spot ÷ distance moved by the solvent front. Spot 1: 1.5 ÷ 7.5 = 0.20. Spot 2: 4.1 ÷ 7.5 = 0.547, which is 0.55 to 2 decimal places. Spot 3: 6.3 ÷ 7.5 = 0.84. Check: each Rf value is between 0 and 1, because no spot can move further than the solvent front.

(b) Match each Rf value with the table: 0.20 is P, 0.55 is R and 0.84 is T. So X contains P, R and T, and does not contain Q or S.

(c) How far a substance moves depends on how soluble it is in the solvent and how strongly it is attracted to the paper. In a different solvent the same colouring moves a different distance, so its Rf value changes. The book's values are for water only, so they cannot be compared with Rf values measured in ethanol.

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16 · Part 1

Paper 1 & 2 Elements, compounds and mixtures [6]

Mark scheme

(a) **[2]**

- ink would dissolve in the solvent OR water and separate into its own dyes (1)
- these dyes would mix with OR be confused with the spots from the black ink, but pencil does not dissolve (1)

(b) **[2]**

- otherwise the black ink would dissolve into the water in the beaker (1)
- instead of moving up the paper, so the dyes would not separate OR no spots would form on the paper (1)

(c) **[1]**

- it is a mixture (of at least four substances) (1)

ALLOW it contains at least three dyes that are soluble in water

REJECT it is a pure substance

(d) **[1]**

- that substance is insoluble in water OR the solvent (1)

Worked solution

(a) The line marks where the ink starts. If it were drawn in ink, the ink of the line would also dissolve and move up the paper, adding extra spots that would mix with the results. Pencil (graphite) does not dissolve in water, so it stays where it was drawn.

(b) If the water level is above the spot, the black ink dissolves straight into the water in the beaker and is lost. The spot must be above the water so that the water soaks up the paper and carries the dyes upwards, separating them.

(c) A pure substance gives only one spot. The black ink gives four spots in total (three that move and one that stays on the line), so it is a mixture of at least four substances.

(d) A substance only moves up the paper if it dissolves in the solvent. The spot that stays on the line contains a substance that is insoluble in water.

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17 · Part 1 Paper 1 & 2 Atomic structure [1]

Answer: D

Explanation

A proton has a relative mass of 1 and a relative charge of +1, so D is correct. A an electron has a negligible relative mass (about 1/1840) and a relative charge of −1. B a neutron has a relative mass of 1 but no charge. C a hydrogen atom has a relative mass of about 1 but no overall charge, because its proton and electron charges cancel.

18 · Part 1 Paper 1 & 2 Atomic structure [1]

Answer: B

Worked solution

Look only at the number of protons first: atoms of the same element have the same number of protons (the same atomic number). W and Y are the only pair with the same number of protons (6). They have different numbers of neutrons, so they are isotopes.

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19 · Part 1 Paper 1 & 2 Atomic structure [8]

Mark scheme

(a) **[3]**

- fluorine-19: 9 protons, 10 neutrons, 9 electrons (1)
- calcium-40: 20 protons, 20 neutrons, 20 electrons (1)
- iron-56: 26 protons, 30 neutrons, 26 electrons (1)

(b) **[1]**

- the total number of protons and neutrons in the nucleus (of an atom) (1)

> **REJECT** number of neutrons only OR mass of the atom

(c) **[2]**

- protons and neutrons are in the nucleus (at the centre of the atom) (1)
- electrons are in shells OR energy levels around the nucleus (1)

(d) **[2]**

- the number of protons equals the number of electrons (1)
- protons have a charge of +1 and electrons a charge of −1, so the charges cancel (1)

Worked solution

(a) The atomic number from the Periodic Table gives the number of protons: F 9, Ca 20, Fe 26. In an atom, the number of electrons equals the number of protons. Number of neutrons = mass number − atomic number: 19 − 9 = 10, 40 − 20 = 20, 56 − 26 = 30.

(b) The mass number counts all the particles in the nucleus: protons plus neutrons.

(c) An atom has a tiny central nucleus containing the protons and neutrons. The electrons are arranged in shells around the nucleus.

(d) Each proton has a relative charge of +1 and each electron −1. An atom has equal numbers of protons and electrons, so the positive and negative charges cancel and the atom is neutral. Neutrons have no charge.

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20 · Part 1 Paper 1 & 2 Atomic structure [7]

Mark scheme

(a) **[2]**

- both have the same number of protons (12) OR the same atomic number (1)
- they have different numbers of neutrons (12 and 14) (1)

(b) **[3]**

- (24 × 79.0) + (25 × 10.0) + (26 × 11.0) OR 1896 + 250 + 286 = 2432 (1)
- divided by 100 (1)
- = 24.3 (1)

REJECT 24.32 for the third mark (the question asks for one decimal place); the first two marks are still awarded

ECF from one incorrect product, for the second and third marks

(c) **[2]**

- relative atomic mass is a (weighted) average OR mean of the masses of the isotopes (1)
- magnesium contains isotopes of different masses in different proportions OR abundances (1)

Worked solution

(a) The element is decided by the number of protons. Both atoms are magnesium, so both have 12 protons. Magnesium-24 has 24 − 12 = 12 neutrons and magnesium-26 has 26 − 12 = 14 neutrons. Same protons, different neutrons means they are isotopes.

(b) Multiply each isotope's mass by its percentage, add, then divide by 100: (24 × 79.0 + 25 × 10.0 + 26 × 11.0) ÷ 100 = (1896 + 250 + 286) ÷ 100 = 2432 ÷ 100 = 24.32, which is 24.3 to one decimal place. Check: the answer is between 24 and 26 and close to 24, because most atoms are magnesium-24.

(c) Each isotope has a whole-number mass, but Ar is a weighted mean of all the atoms in the sample. Because the sample has a mixture of isotopes with different masses, the mean is not a whole number.

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21 · Part 1 Paper 1 & 2 Atomic structure [7]

Mark scheme

(a) **[1]**

- two or more atoms chemically bonded OR joined together (1)

ALLOW two or more atoms joined by covalent bonds

(b) **[1]**

- boron-10 has one fewer neutron (5 instead of 6) (1)

ALLOW boron-10 has a different number of neutrons OR a smaller mass number

REJECT different number of protons OR electrons

(c) **[2]**

- 10.8 is closer to 11 than to 10 (1)
- relative atomic mass is a (weighted) mean, so it is closer to the mass of the more abundant isotope (1)

(d) **[3]**

- let the percentage of boron-10 be x: $10x + 11(100 - x) = 10.8 \times 100$ (1)
- $1100 - x = 1080$ (1)
- $x = 20\%$ (1)

ALLOW other correct methods, e.g. $(11 - 10.8) \div (11 - 10) \times 100 = 20\%$, for 3 marks

ALLOW trial with 20% boron-10 and 80% boron-11 shown to give 10.8, for 3 marks

Worked solution

(a) A molecule is two or more atoms chemically bonded together, such as one boron atom bonded to three fluorine atoms in BF_3.

(b) Isotopes of an element have the same number of protons (boron has 5) but different numbers of neutrons. Boron-10 has $10 - 5 = 5$ neutrons and boron-11 has $11 - 5 = 6$ neutrons.

(c) If the two isotopes were equally common, Ar would be 10.5. The value 10.8 is nearer to 11, so the weighted mean is pulled towards the mass of boron-11, which means boron-11 atoms are more common.

(d) Let x be the percentage of boron-10, so boron-11 is $(100 - x)$. Then $(10x + 11(100 - x)) \div 100 = 10.8$. So $10x + 1100 - 11x = 1080$, giving $1100 - x = 1080$ and $x = 20$. The sample is 20% boron-10. Check: $(10 \times 20 + 11 \times 80) \div 100 = (200 + 880) \div 100 = 10.8$.

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22 · Part 1 Paper 1 & 2 The Periodic Table [1]

Answer: C

Explanation

The modern Periodic Table arranges elements in order of increasing atomic number (number of protons), so C is correct. A and D are wrong because atoms of one element can have different numbers of neutrons and different mass numbers (isotopes). B is wrong because a few pairs of elements are out of order by relative atomic mass, for example argon (40) comes before potassium (39).

23 · Part 1 Paper 1 & 2 The Periodic Table [1]

Answer: A

Worked solution

For the main groups, the period number equals the number of occupied shells and the group number equals the number of outer-shell electrons. Fill the shells 2, then 8, then put 6 in the third shell: 2.8.6.

24 · Part 1 Paper 1 & 2 The Periodic Table [1]

Answer: D

Explanation

Non-metals are found on the right of the Periodic Table. The element in Group 7, Period 2 is fluorine, a non-metal, so D is correct. A (potassium), B (magnesium) and C (aluminium) are all on the left-hand side or in the metal region, so they are metals.

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25 · Part 1 Paper 1 & 2 The Periodic Table [6]

Mark scheme

(a) **[4]**

- nitrogen: 2.5 (1)
- aluminium: 2.8.3 (1)
- chlorine: 2.8.7 (1)
- calcium: 2.8.8.2 (1)

ALLOW commas instead of dots, e.g. 2,8,7

(b) **[2]**

- chlorine has 7 electrons in its outer shell, so it is in Group 7 (1)
- chlorine has 3 occupied shells, so it is in Period 3 (1)

Worked solution

(a) Fill the shells in order: the first shell holds up to 2 electrons, the second up to 8, and for the first 20 elements the third holds up to 8 before electrons go into the fourth. Nitrogen (7): 2.5. Aluminium (13): 2.8.3. Chlorine (17): 2.8.7. Calcium (20): 2.8.8.2. Check that the numbers add up to the atomic number each time.

(b) The number of electrons in the outer shell gives the group number (7 for chlorine). The number of occupied shells gives the period number (3 for chlorine).

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26 · Part 1 Paper 1 & 2 The Periodic Table [6]

Mark scheme

(a) **[2]**

- Group 5 (1)
- Period 3 (1)

(b) **[2]**

- E and G (1)
- both have 2 electrons in the outer shell (1)

(c) **[1]**

- J and L (1)
- both needed

(d) **[1]**

- lithium (1)

Worked solution

(a) J is 2.8.5: three occupied shells means Period 3, and five outer electrons means Group 5.

(b) Elements in the same group have the same number of outer-shell electrons. E (2.2) and G (2.8.2) both have 2 outer electrons, so both are in Group 2.

(c) Work out each position. E: Group 2, Period 2. G: Group 2, Period 3. M: Group 1, Period 2. These are on the left of the Periodic Table, so they are metals. L (2.4) is in Group 4, Period 2 and J (2.8.5) is in Group 5, Period 3; both are on the right-hand side, so they are non-metals.

(d) M has 2 + 1 = 3 electrons, so its atomic number is 3. Element 3 in the Periodic Table is lithium.

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27 · Part 1 Paper 1 & 2 The Periodic Table [6]

Mark scheme

(a) **[2]**

- X is a metal, because it is a good conductor and its oxide is basic OR forms an alkaline solution (pH 11) (1)
- Y is a non-metal, because it is a poor conductor and its oxide is acidic OR forms an acidic solution (pH 2) (1)

(b) **[2]**

- Z is a good conductor of electricity, which is typical of metals, so a student might classify Z as a metal (1)

> **ALLOW** Z is (a form of) carbon OR graphite, a non-metal that conducts electricity, for the first mark

- its oxide is acidic OR forms a solution with pH 5 OR below 7, and acidic oxides are formed by non-metals (1)

> **REJECT** the oxide is basic OR alkaline

(c) **[2]**

- add (dilute) acid, e.g. dilute hydrochloric acid OR dilute sulfuric acid, to the copper(II) oxide (1)
- the copper(II) oxide reacts OR dissolves OR neutralises the acid, forming a (blue) solution of a salt (and water) (1)

> **REJECT** add an alkali OR add water and use indicator

Worked solution

(a) Metals are good conductors and form basic oxides; a basic oxide that dissolves in water gives an alkaline solution (pH above 7). X has both properties. Non-metals are usually poor conductors and form acidic oxides, which give solutions with pH below 7. Y has both properties.

(b) Z conducts electricity, which on its own suggests a metal, so a student using only the conductivity result would classify Z wrongly. But its oxide gives a solution of pH 5, which is acidic, and acidic oxides are formed by non-metals. Some non-metals, such as graphite (a form of carbon), conduct electricity, so the oxide result is the one that shows Z is a non-metal.

(c) A basic oxide reacts with an acid to form a salt and water. Add dilute acid to the black copper(II) oxide and warm. The solid dissolves and the solution turns blue, because a copper(II) salt forms. This shows that copper(II) oxide is basic, even though it is insoluble in water.

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28 · Part 1 Paper 1 & 2 The Periodic Table [6]

Mark scheme

(a) **[2]**

- they all have one electron in the outer shell (1)
- chemical properties depend on the number of outer-shell electrons OR they all react by losing one electron (1)

(b) **[2]**

- rubidium hydroxide (1)
- hydrogen (1)

(c) **[2]**

- sodium is 2.8.1 and silicon is 2.8.4 (1)
- sodium has one outer electron and silicon has four; the group number equals the number of outer-shell electrons, so they are in different groups (1)

ALLOW silicon is in Group 4 and sodium in Group 1, for the second mark if linked to outer electrons

Worked solution

(a) Lithium is 2.1, sodium is 2.8.1 and potassium is 2.8.8.1. Each has just one electron in its outer shell. Reactions involve the outer electrons, so elements with the same number of outer electrons react in similar ways.

(b) Rubidium is in Group 1, so it also has one outer electron and reacts like sodium. Sodium gives sodium hydroxide and hydrogen, so rubidium gives rubidium hydroxide and hydrogen.

(c) Sodium (11 electrons) is 2.8.1 and silicon (14 electrons) is 2.8.4. Both have three occupied shells, so both are in Period 3. Sodium has one outer electron (Group 1) and silicon has four (Group 4), so they are in different groups.

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29 · Part 1 Paper 1 & 2 The Periodic Table [6]

Mark scheme

(a) **[2]**

- neon: 2.8 (1)
- argon: 2.8.8 (1)

(b) **[2]**

- their atoms have full outer shells OR stable electronic configurations (1)
- so they do not readily lose, gain or share electrons (1)

REJECT they have no electrons in the outer shell

(c) **[2]**

- helium's outer shell is full, because the first shell can hold only 2 electrons (1)
- so, like the other noble gases, helium is unreactive OR does not readily lose, gain or share electrons (1)

Worked solution

(a) Neon (atomic number 10): 2.8. Argon (atomic number 18): 2.8.8.

(b) The atoms of the noble gases already have full outer shells, which is a stable arrangement. They have no tendency to lose, gain or share electrons, so they do not readily form compounds.

(c) Helium has only one shell, and the first shell is full with 2 electrons. So helium has a full outer shell, just like neon and argon, and it is also unreactive. Its properties match Group 0, so it is placed there rather than in Group 2.

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30 · Part 1 Paper 1 & 2 The Periodic Table [7]

Mark scheme

(a) **[2]**

- Group 2 (1)
- Period 5 (1)

(b) **[1]**

- 2 (1)

ECF from the group given in (a)

(c) **[2]**

- good conductor (1)
- strontium is a metal because it is on the left of the Periodic Table OR in Group 2 (1)

(d) **[2]**

- strontium chloride and water (1)
- strontium oxide is a basic oxide (1)

Worked solution

(a) Find element 38 in the Periodic Table. It is in the same column as magnesium and calcium (Group 2) and in the fifth row (Period 5).

(b) For a main-group element, the group number equals the number of outer-shell electrons, so strontium has 2.

(c) Strontium is on the left-hand side of the Periodic Table, in Group 2, so it is a metal. Metals are good conductors of electricity.

(d) Strontium is in the same group as calcium, so their compounds react in similar ways. Calcium oxide + hydrochloric acid gives calcium chloride + water, so strontium oxide + hydrochloric acid gives strontium chloride + water. An oxide that reacts with an acid to form a salt and water is a basic oxide, which is typical of a metal.

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